

Fiducial-Interleaved Count the Bucket

(FICTB)

A method for phase-alignment of distributed digital systems.

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Overview

Count the Bucket is a common method for precision timing in beam-synchronous RF systems. Periodic incorporation of a global fiducial allows for local nodes to digitally align their bucket counts to the reference node, and execute inter-bucket triggers without phase matching of the local RF. This fiducial is provided by a 'Pulse Per Second' signal that occurs simultaneously for all connected nodes. The 'Fiducial-Interleaved Count the Bucket' (FICTB) method is described herein.

System Description

FICTB relies on two inputs: PPS, and an RF Clock. Additionally, a subharmonic of the RF Clock called the 'Frame' or 'Event Window' is utilized to allow jitter in packet transmission that may exceed the local RF period.

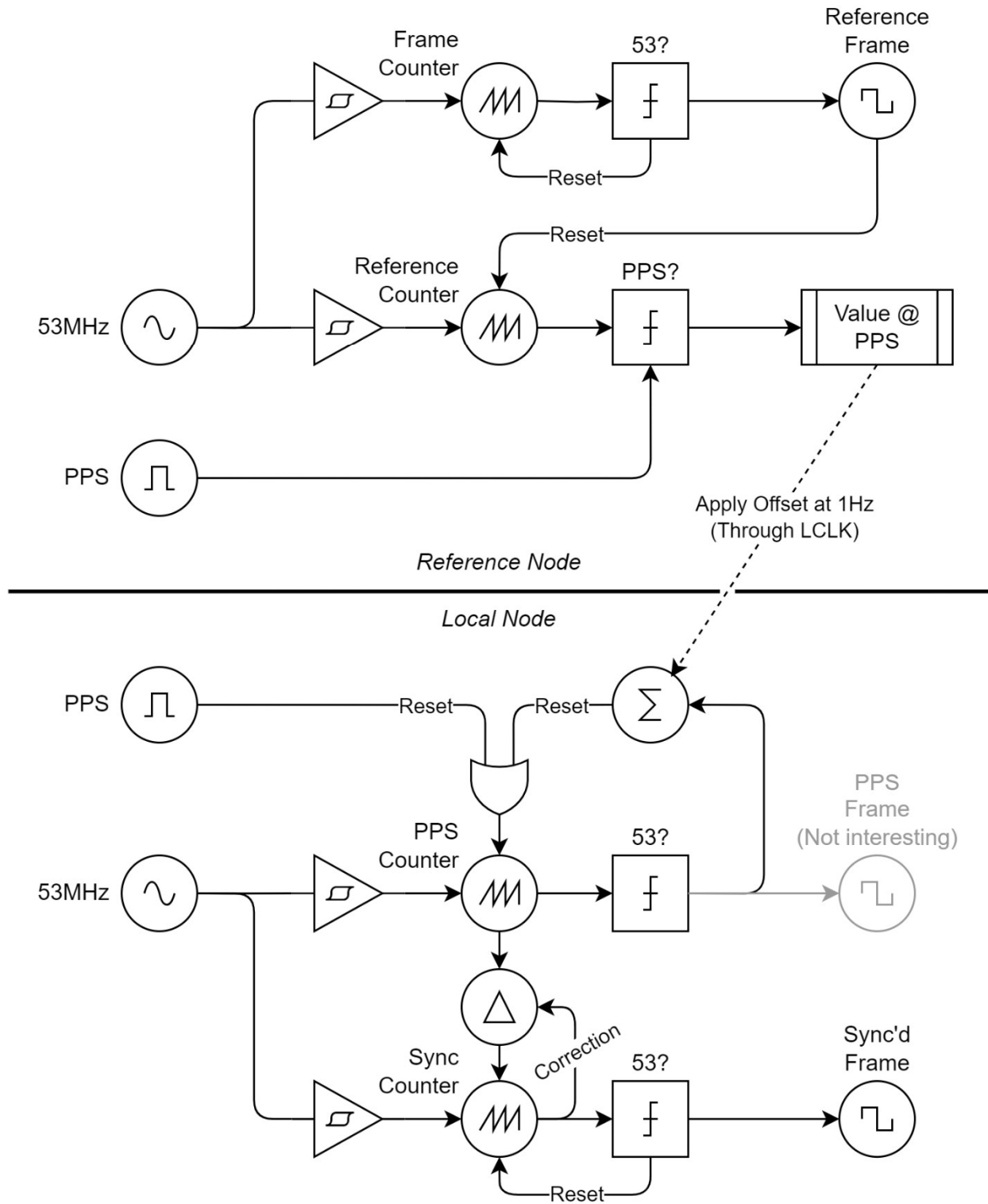
The reference node is responsible for Frame tracking. At each Pulse Per Second, it captures the number of RF buckets that have elapsed from the frame edge to the PPS in order to determine the offset between the RF phase, and the fiducial. This information is transmitted to downstream nodes where they can apply their own offsets and 'lock-in' to the global timing system.

Local nodes also perform frame tracking against the PPS. Ideally this remains constant after the first lock, but variations in RF and network conditions may induce edge cases. To compensate, the 'Sync Frame' utilizes a feedback loop referenced against the PPS frame to provide stable, RF-synchronous event windows at local devices.

Finally, events are received and synchronized to the local Sync Frame. Capturing events in a fixed window allows for a high degree of long-term stability in the timing system.

Please reference the diagram below.

FICTB Implementation



FPGA Modules

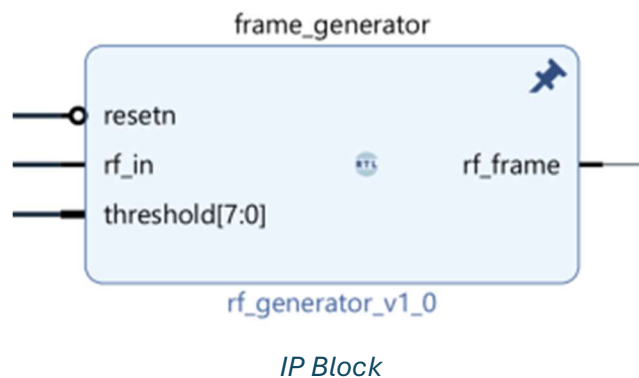
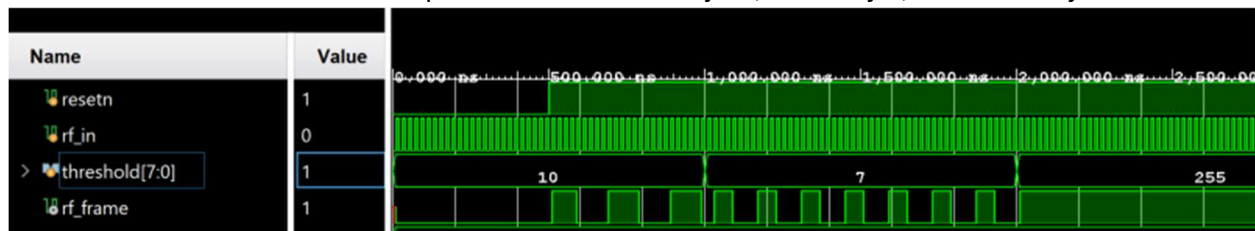
Frame Generator

This module takes an RF clock input (162.5MHz) and generates a square wave at the RF subharmonic determined by the threshold (divide by 16 = 10.16MHz). The resulting 'frame' is the period to which Clock (LCLK) events align. This frame may also be referred to as an 'Event Window'.

The Frame Generator IP block is only required at the LCLK Generator, downstream frames will be sync'd according to the 'phase offset event' that transmit the RF phase relative to a Pulse Per Second (PPS), as determined by the reference node.

Frame Generator Implementation

The Frame Generator creates a square wave for divide by 10, divide by 7, and divide by 255:



Frame Generator Notes

The Frame Generator may be adapted to any RF subharmonic, such as Main Injector's $53\text{MHz} \div 7$ that determines the 7.5MHz beam synchronous timing system.

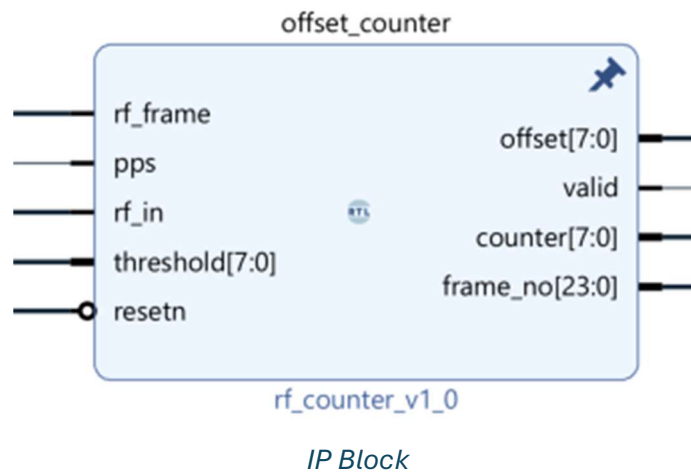
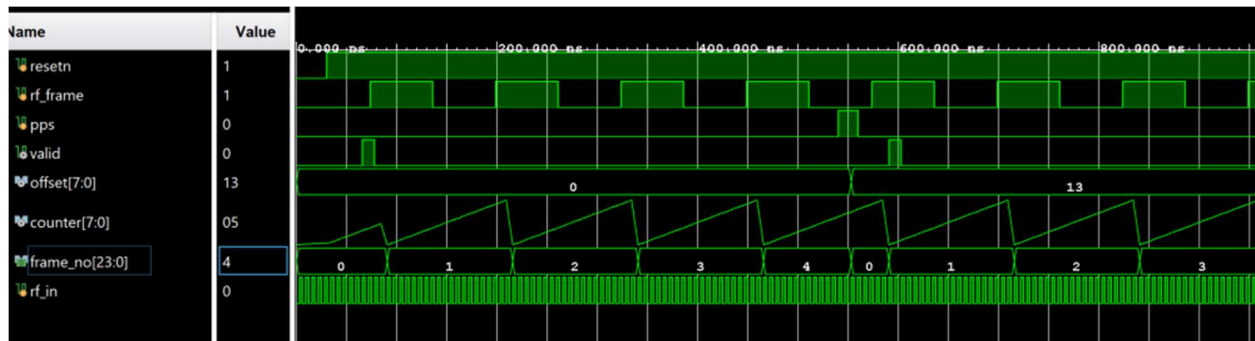
The LCLK generator does not necessarily need the frame generator, the 10.16MHz subharmonic may be provided from an external source. It *may* be beneficial to use a subharmonic tied directly to the master oscillator.

Offset Counter

The Offset Counter takes an RF clock input, the RF Frame, and Pulse Per Second in order to determine the frame Offset relative to the PPS signal. The Offset Counter is only required at the LCLK generator (reference), downstream systems will align their local frames to the phase information provided by the reference offset.

Offset Counter Implementation

The offset counter updates its offset at PPS and resets the frame number. When valid is asserted, these numbers are ready to be transmitted to downstream systems.



Offset Counter Notes

PPS and RF frame signals are expected to be asynchronous. A triple debounce is utilized (pps_1, pps_2, pps_3 etc.) to ensure all counts are accurate cycle-by-cycle. An additional 'Frame Counter' provides the index of the frame relative to the PPS.

'Valid Counter' provides a delay to the strobe that indicates the offset is valid. This was implemented for testing purposes and is not a strict requirement of the module.

'Threshold' was also used for testing purposes and is not a strict requirement of the module.

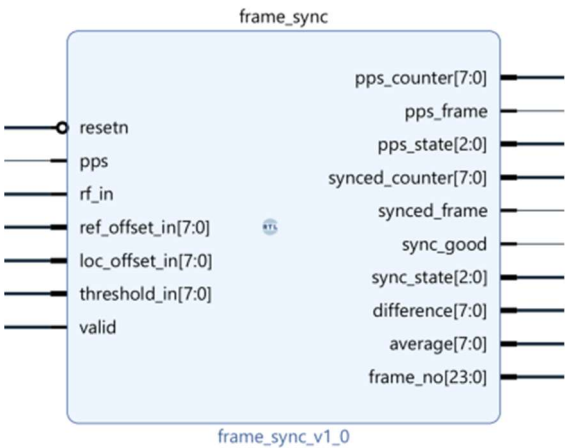
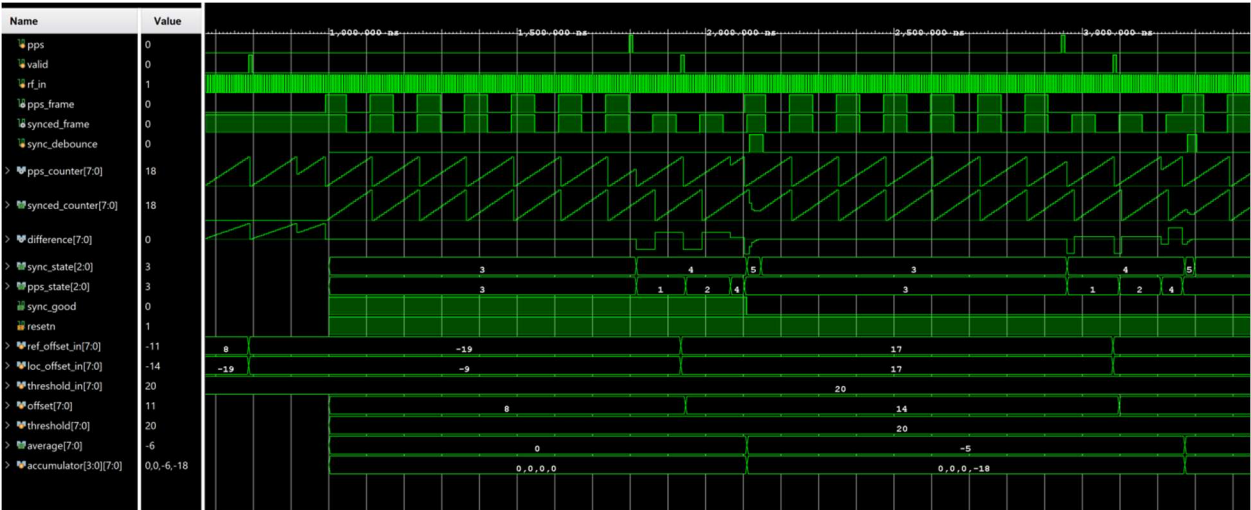
Frame Sync

The Frame Sync module synchronizes the phase of local events to the LCLK generator. This allows for local triggers to happen synchronously with the beam. This module provides a Synced Frame, A Synced Counter, and Sync Good signals to indicate when it has locked to the reference node.

The Frame Sync module has inputs for Local Offset and Reference Offset. The Local Offset allows users to statically adjust the node relative to the immediate RF. The Reference Offset is the phase information from the LCLK generator, and syncs the device relative to the reference RF.

Frame Sync Implementation

Frame Sync contains two state machines. One for tracking its local frame against the PPS, and one for re-aligning the synced frame when it falls out of tolerance. The ‘sync’ state machine runs a simple feedback loop to reduce jitter during PPS~RF edge cases.



Frame Sync IP Module

Frame Sync Notes

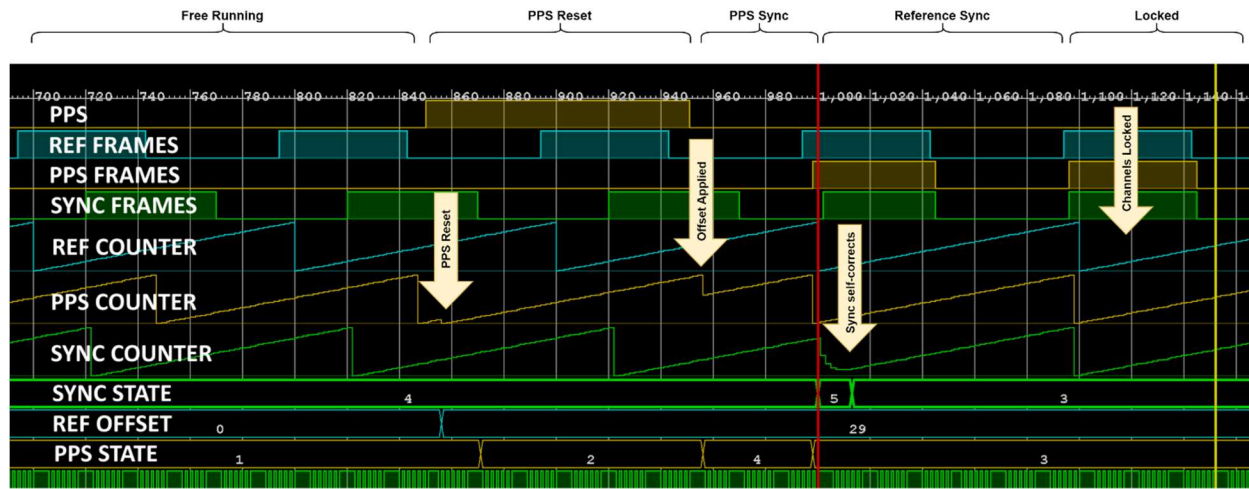
The PPS counter maintains its own 'PPS frame', which is the same subharmonic as the reference frame, except it resets after each PPS. This provides an updated phase measurement for the sync frame at each cycle. The PPS counter runs through 5 states:

- **Initialize**
Happens on reset
- **Ready for Offset**
After the PPS signal, waits for the Reference offset event from the LCLK generator, and calculates the local offset to be applied.
- **Ready for Reset**
Waits until the next PPS frame edge, and then apply the local offset (waiting for the PPS frame prevents rapid fluctuations).
- **Syncing**
Waits one PPS frame to allow things to settle.
- **Locked**
Continues generating the PPS frame, now in sync with the LCLK reference. When PPS is received, it returns to 'Ready for Offset.'

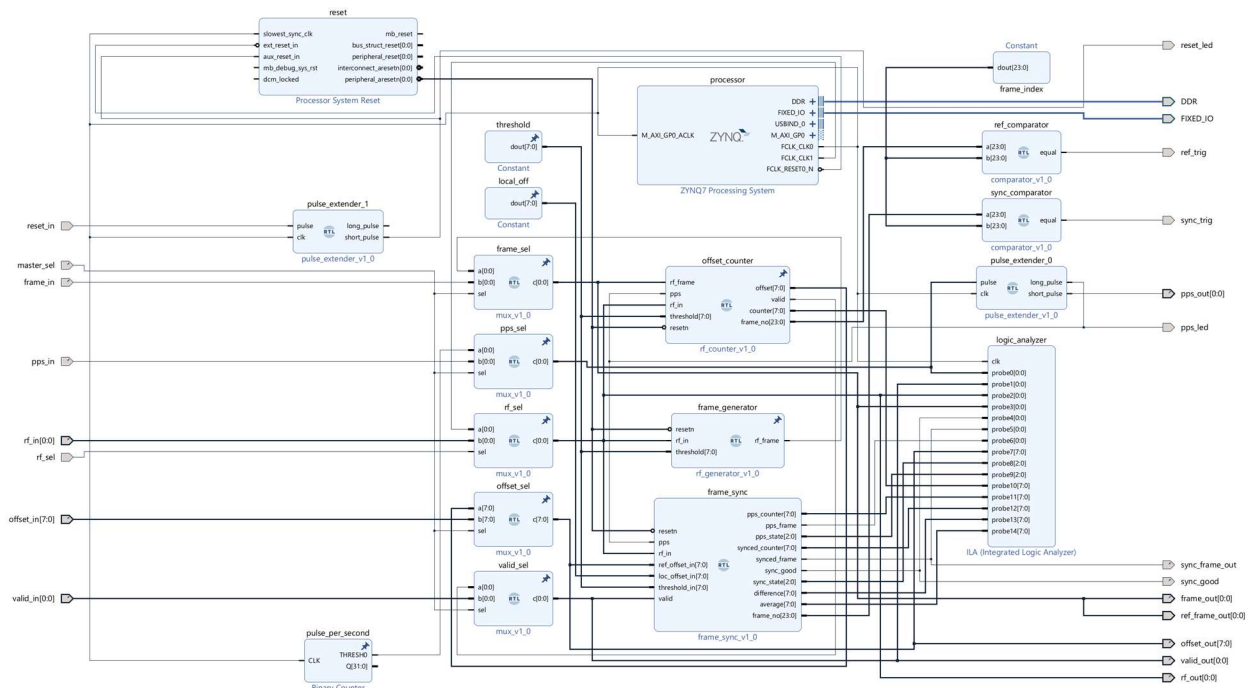
The Sync counter maintains the actual event window frame to be utilized by the node. This frame runs continuously and only adjusts when major phase errors are observed. A 'rubber banding' algorithm is used to prevent frames from changing too rapidly. This was added for fun and may not be strictly necessary. The Sync counter has 4 states:

- **Initialize**
Happens on reset
- **Locked**
Generates sync'd frame until a PPS is received, at which point it re-evaluates.
- **Syncing**
Once the PPS frame is synced, this state updates the accumulator that tracks the error.
- **Adjusting**
If the error is greater than one bucket, it gradually applies correction until the sync frame is aligned with the reference frame, and returns to 'locked'.

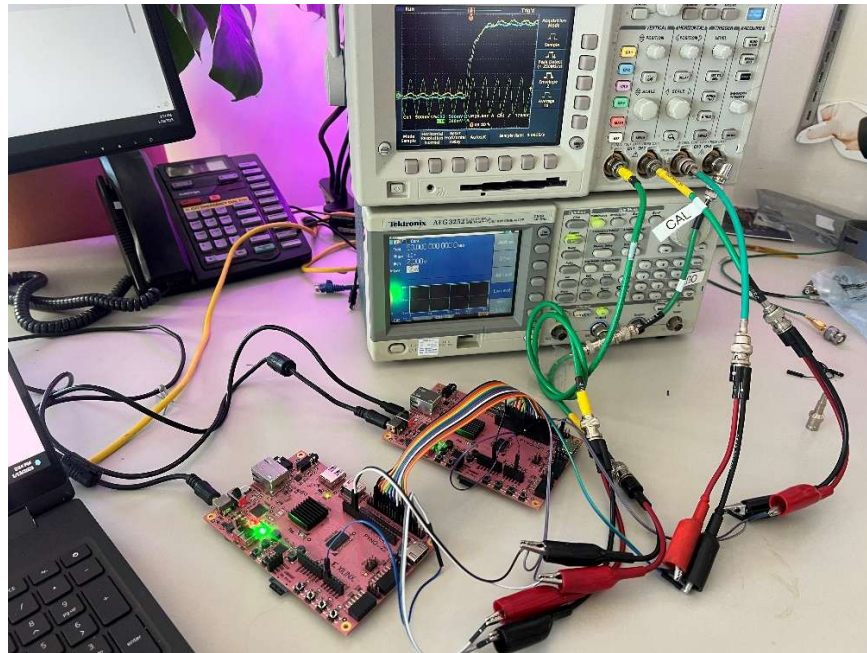
FICTB Update 1/13/25



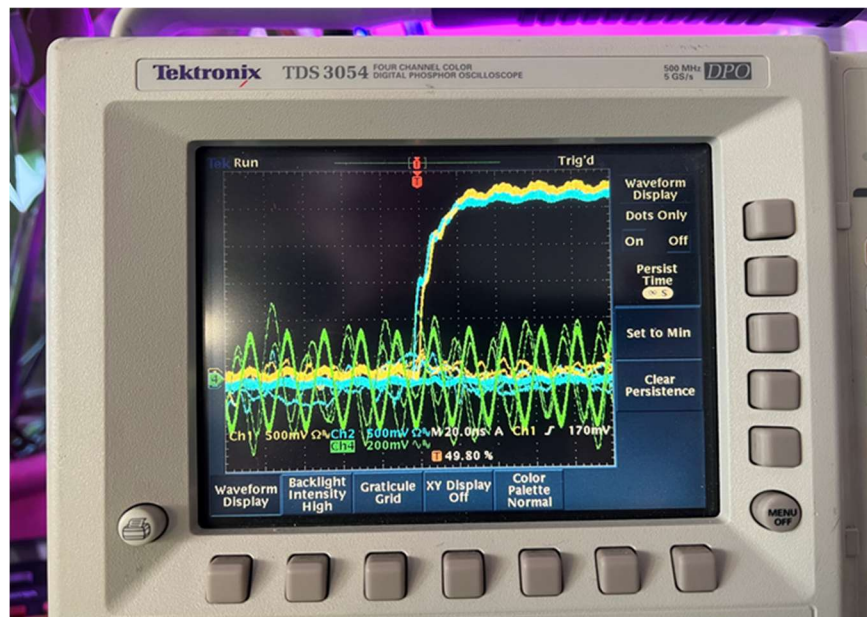
FICTB Results



FICTB Block Design



Test setup



Jitter measurement after one hour

Used PYNQ Z-2 (Zynq 7000)

Bit banging offset to second unit

Locks within 10s

No jitter after 1hr

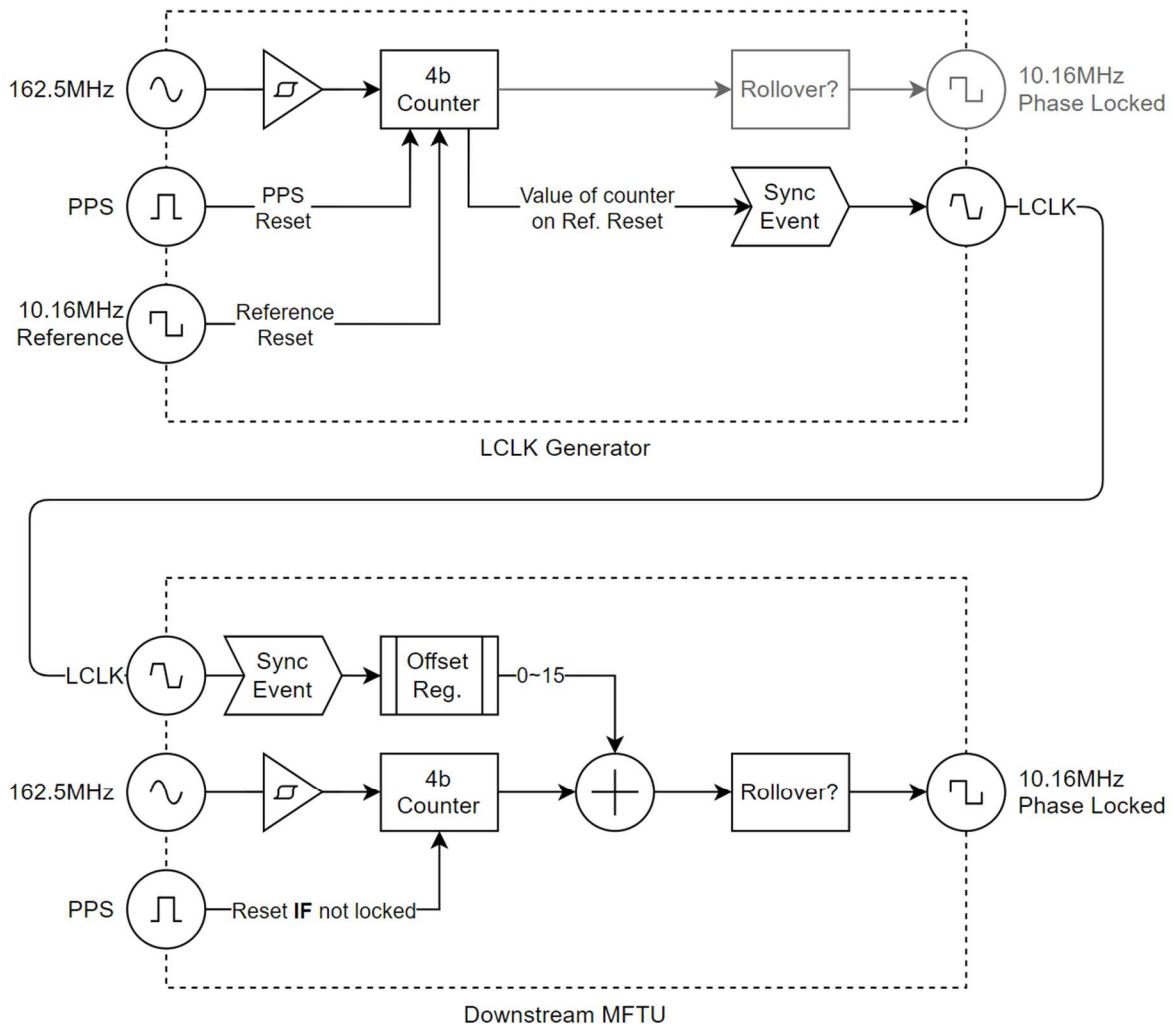
Utilizing 53MHz/ 1MHz frames~ PYNQ terminations won't handle 162.5MHz

Works w/ internal clock at 166.66MHz

Does not meet timing

LCLK Phase Alignment Test Plan (Dec 2024)

- 1) Generator counts buckets between PPS and 10MHz reference frame
- 2) This is broadcast as a 'sync event', with an offset from 0~15
- 3) The downstream device applies the offset to its 4b counter
- 4) The phase-locked 10MHz occurs whenever the counter + offset rolls over
- 5) This can be broadcast at 1Hz, periodically re-synced.



Notes:

Downstream devices will want a second 0~15 offset register to phase align locally

PPS and 162.5 are asynchronous clocks, will never align perfectly.

Unlikely LCLK packets will arrive perfectly between phase locked windows, keep an eye on jitter.

1.3Gbps / 128 = 10.16MHz ... 128b LCLK packets? any reason for Manchester?

$162.5 / 16 = 10.16\text{MHz}$